sPHENIX Internal Note

Photon Efficiency and Energy Response with ΔR Truth-Matching Removed: Single, Double, and Mixed Interaction MC

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

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Abstract

Building on the companion study [1] that demonstrated the $\Delta R < 0.1$ truth-matching cut rejects ∼3.5% of correctly track-ID-matched truth photons in single-interaction MC, this note quantifies the efficiency impact of removing the cut across three MC sample categories: single-interaction, double-interaction, and physics-mixed (0 mrad, 77.6%/22.4%). The factorized efficiency chain $\varepsilon_{\text{reco}} \to \varepsilon_{\text{iso|reco}} \to \varepsilon_{\text{id|reco+iso}} \to \varepsilon_{\text{all}}$ is computed with and without the ΔR requirement in a single pass over `photon10` samples. Removing the ΔR cut improves the total efficiency ε_{all} by +2.1% (single), +13.8% (mixed), and approximately doubles it for double-interaction MC. The reconstruction stage dominates the change; isolation efficiency is unaffected; photon ID (BDT) efficiency shows a measurable degradation in double-interaction MC from vertex-shifted cluster kinematics. The single/double/mixed overlay provides a clean decomposition of pileup effects on each efficiency stage.

We then extend the study to the photon *energy response* $R_{E_{\text{T}}} \equiv E_{\text{T}}^{\text{reco}}/p_{\text{T}}^{\text{truth}}$ across six sample configurations: single, double, and four mixed variants covering the two RHIC crossing-angle running periods (0 mrad with 22.4% cluster-weighted double-interaction fraction and 1.5 mrad with 7.9%), each with and without the data-driven z-vertex reweighting used in the main pipeline. The response is examined at the trkID-matched ("matched"), post-common-cuts ("reco"), and fully-selected ("selected") levels. At the matched level, the pileup-induced vertex shift produces a direct kinematic bias (∼7 pp mean shift and ∼50% relative width increase comparing double to single); after the common and tight selections the bias is largely absorbed by the cut boundaries, leaving a residual ∼1 pp mean shift and ∼20% width increase. The 1.5 mrad mixed sample (the nominal production period) tracks single-interaction MC closely; the 0 mrad mixed sample exhibits a measurably different response that motivates treating the two running periods separately in the pileup systematic. Vertex reweighting modifies the response by less than ∼1 pp at the reco level, confirming that it is sub-dominant relative to the interaction-type composition itself.

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1 Introduction

The reconstruction efficiency for the sPHENIX isolated photon cross-section measurement is computed from PYTHIA + GEANT4 MC using a factorized chain of selection stages: reconstruction, isolation, and photon identification (BDT). The truth association between generated photons and reconstructed EMCal clusters is established by the GEANT track ID (`trkID`), optionally supplemented by a geometric proximity requirement $\Delta R < 0.1$.

A companion study [1] demonstrated that this ΔR cut is a vertex-resolution artifact: in single-interaction MC it rejects ∼3.5% of `trkID`-matched photons, and in double-interaction MC the failure rate reaches 65%, driven entirely by the vertex displacement $|\Delta z| > 9.35$ cm (the geometric critical threshold set by the EMCal barrel radius). The rejected clusters are well-reconstructed photons with $E_{\mathrm{T}}/p_{\mathrm{T}}^{\mathrm{truth}} \in [0.8, 1.2]$ for 73% of cases.

Based on those findings, the $\Delta R < 0.1$ requirement was removed from the production efficiency calculation. This note quantifies the resulting efficiency changes across all interaction types and examines the interplay between the ΔR removal and the photon ID (BDT) selection, which uses an E_{T}-dependent threshold susceptible to vertex-induced kinematic shifts.

With the ΔR artifact out of the way, Section 6 extends the study to the photon *energy response* $R_{E_{\mathrm{T}}} = E_{\mathrm{T}}^{\mathrm{reco}}/p_{\mathrm{T}}^{\mathrm{truth}}$. The response directly drives the cluster E_{T} used in the ABCD region classification, the E_{T}-dependent BDT threshold, and the unfolding response matrix, so any pileup-induced bias in $R_{E_{\mathrm{T}}}$ feeds straight into the cross-section. Six sample configurations are studied (single, double, and four mixed variants covering the 0 mrad / 1.5 mrad running periods with and without data-driven z-vertex reweighting), at three selection levels that mirror the factorized efficiency chain. This allows us to disentangle interaction-type composition from residual data–MC vertex mismatch — the two independent knobs available to the pileup systematic.

2 Samples and Method

2.1 Monte Carlo Samples

Three sample configurations are studied, all using the `photon10` generator (truth p_{T} range 14–30 GeV):

- **Single-interaction:** `photon10` nominal MC (∼10M events). One *pp* collision per bunch crossing.
- **Double-interaction:** `photon10_double` MC (∼10M events). Full GEANT simulation of two overlapping *pp* collisions per bunch crossing, with the reconstructed vertex at the average of the two collision vertices.
- **Mixed (0 mrad):** Physics-weighted combination of 77.6% single + 22.4% double, corresponding to the cluster-weighted double-interaction fraction computed by `calc_pileup_range.C` for the 0 mrad crossing-angle running period at RHIC.

2.2 Efficiency Definitions

The efficiency is factorized into four stages, each conditional on the previous:

$$\varepsilon_{\text{reco}} = \frac{N(\text{reco + common cuts})}{N(\text{truth photons})} \tag{1}$$

$$\varepsilon_{\text{iso|reco}} = \frac{N(\text{reco + common + isolated})}{N(\text{reco + common})} \tag{2}$$

$$\varepsilon_{\text{id|reco+iso}} = \frac{N(\text{reco + common + iso + tight})}{N(\text{reco + common + iso})} \tag{3}$$

$$\varepsilon_{\text{all}} = \varepsilon_{\text{reco}} \times \varepsilon_{\text{iso|reco}} \times \varepsilon_{\text{id|reco+iso}} = \frac{N(\text{all cuts})}{N(\text{truth photons})} \tag{4}$$

The "common cuts" include cluster probability, shower-shape quality gates (e11/e33, ω_r, ω_η), and optionally the NPB score cut. "Isolated" applies the parametric topo-cluster isolation cut $E_{\text{T}}^{\text{iso}} < b + s \cdot E_{\text{T}}$ (topo $R_{\text{cone}} = 0.4$). "Tight" applies the full shower-shape selection plus the E_{T}-dependent BDT threshold $\text{BDT}_{\text{min}} = \text{intercept} + \text{slope} \times E_{\text{T}}$.

2.3 Matching Modes

Each truth photon is evaluated under two matching modes simultaneously in a single pass:

`withDR`: Requires both `trkID` match *and* $\Delta R < 0.1$ (the legacy requirement).

`noDR`: Requires only the `trkID` match (the new baseline).

Clusters that enter the `noDR` numerator but not the `withDR` numerator are labeled "new clusters"—these are the photons recovered by removing the geometric cut. Their BDT pass rate is tracked separately.

The analysis macro is `compare_efficiency_deltaR.C`, which writes all `TEfficiency` objects to a single output ROOT file.

3 Results: Per-Stage Efficiency Comparison

Table 1 presents the p_{T}-integrated efficiency at each factorized stage for all three sample sets, computed with and without the $\Delta R < 0.1$ requirement.

Table 1: Integrated efficiency at each factorized stage for single-interaction, double-interaction, and mixed (0 mrad) MC. "withDR" denotes the legacy matching; "noDR" denotes `trkID`-only matching.

	Single		**Double**		**Mixed 0 mrad**	
Stage	withDR	noDR	withDR	noDR	withDR	noDR
$\varepsilon_{\text{reco}}$	91.4%	93.5%	30.7%	73.1%	73.4%	87.5%
$\varepsilon_{\text{iso\|reco}}$	74.2%	74.2%	71.1%	71.0%	73.8%	73.4%
$\varepsilon_{\text{id\|reco+iso}}$	82.9%	82.7%	82.6%	69.3%	82.9%	79.5%
ε_{all}	56.2%	57.4%	18.0%	36.0%	44.9%	51.1%

Table 2 shows the ratio of `noDR` to `withDR` efficiency at each stage, quantifying the impact of removing the cut.

The key observations are:

Table 2: Ratio of efficiency without the ΔR cut to efficiency with the cut ($\varepsilon_{\text{noDR}}/\varepsilon_{\text{withDR}}$) at each stage. Values above 1.0 indicate improvement from cut removal; values below 1.0 indicate degradation in the recovered cluster population.

Stage	**Single**	**Double**	**Mixed 0 mrad**
$\varepsilon_{\text{reco}}$	1.023	2.386	1.191
$\varepsilon_{\text{iso}\|\text{reco}}$	1.000	0.998	0.994
$\varepsilon_{\text{id}\|\text{reco+iso}}$	0.997	0.840	0.960
ε_{all}	1.021	1.999	1.137

- **Reconstruction** ($\varepsilon_{\text{reco}}$): This stage captures the full impact of the ΔR removal. Single MC improves by 2.3%, reflecting the small population of vertex-displaced clusters. Double MC improves by a factor of ∼2.4, recovering the 65% of clusters that failed the ΔR cut due to vertex displacement. The residual 27% gap between double and single noDR efficiency reflects genuine missing clusters (no `trkID` match at all: 2.4% for single, 7.9% for double).
- **Isolation** ($\varepsilon_{\text{iso}|\text{reco}}$): Essentially unchanged across all samples (ratios within 0.5% of unity). The isolation requirement tests calorimeter energy in an annular cone around the cluster; vertex displacement does not significantly alter the energy sum, and the extra energy from the second interaction causes only a ∼3% absolute degradation comparing single to double.
- **Photon ID** ($\varepsilon_{\text{id}|\text{reco+iso}}$): Single MC shows negligible change (0.2% decrease). Double MC shows a significant 16% decrease (ratio 0.840), indicating that the recovered clusters have measurably worse BDT pass rates. Mixed MC interpolates at 4.0% decrease.
- **Total** (ε_{all}): Net improvement in all cases. The reconstruction gain dominates the BDT degradation. Single: +2.1%; mixed: +13.8%; double: nearly doubling from 18.0% to 36.0%.

Figure 1 shows the p_{T}-dependent efficiency ratio for all four stages and all three sample sets.

4 Overlay: Efficiency without ΔR across Interaction Types

With the ΔR cut removed, the remaining efficiency differences between single, double, and mixed MC arise from genuine pileup physics effects rather than truth-matching artifacts. Figure 2 presents the four-panel overlay of the `noDR` efficiency for all three sample types, providing a clean decomposition of pileup degradation at each selection stage.

4.1 Reconstruction Efficiency

Single-interaction MC achieves $\varepsilon_{\text{reco}} = 93.5\%$; double-interaction drops to 73.1%, a 20 percentage point (pp) reduction. The p_{T} dependence is weak: single MC is flat near 93–95% across the full range, while double MC spans 70–77%, maintaining a roughly constant ∼20 pp gap. This gap reflects two effects:

1. **Genuine cluster loss:** In double-interaction events, the additional hadronic activity can merge the signal photon cluster with nearby energy deposits, causing the `trkID` to be lost from the leading cluster. The no-`trkID`-match rate is 7.9% for double MC vs. 2.4% for single MC, accounting for ∼5.5 pp of the gap.

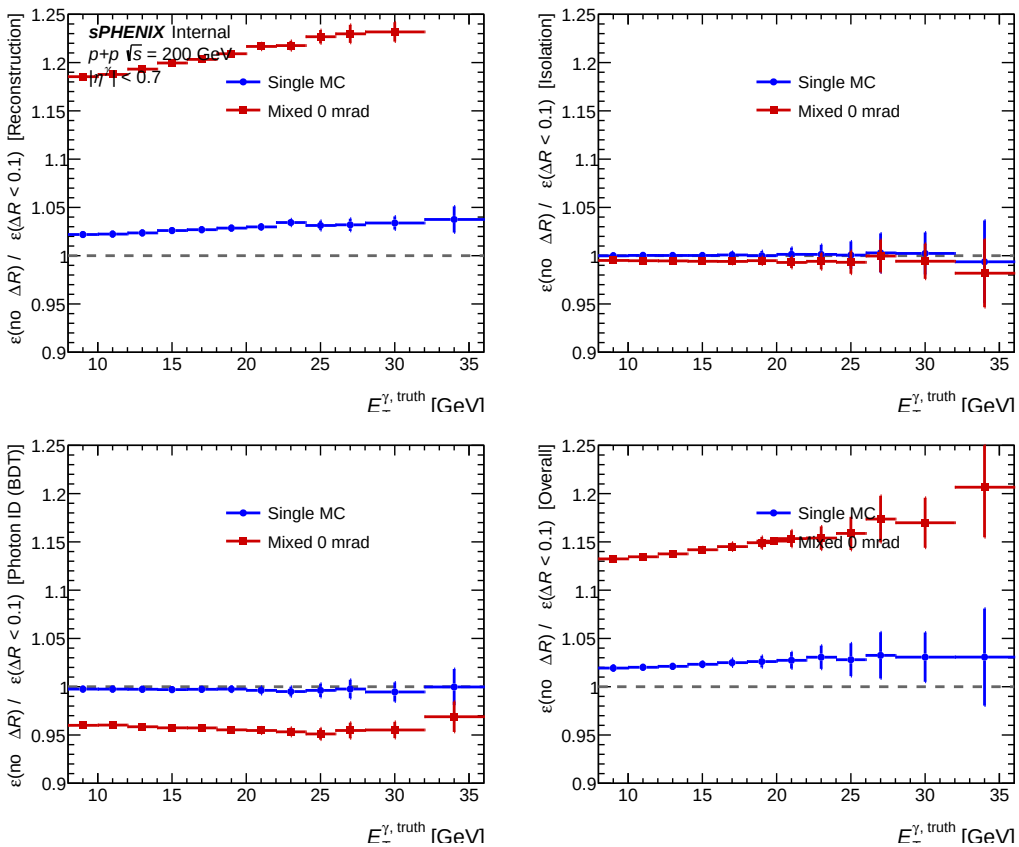

Figure 1: Ratio of efficiency without ΔR cut to efficiency with ΔR cut as a function of truth p_{T} for single (blue) and mixed (red) MC. The reconstruction stage shows the largest change, particularly for the mixed sample containing 22.4% double-interaction events. Isolation is flat. Photon ID shows a deficit in the mixed sample at all p_{T}.

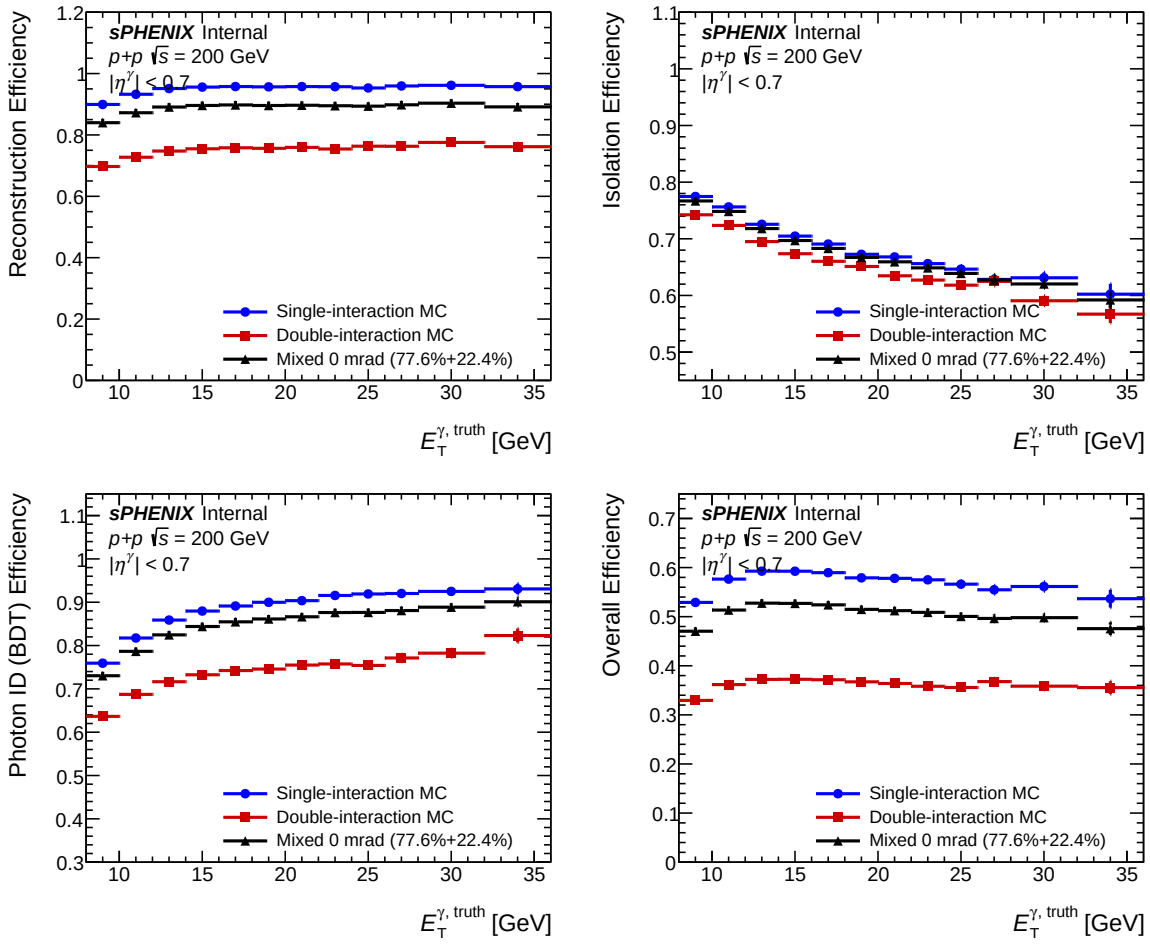

Figure 2: Per-stage efficiency as a function of truth p_{T} without the ΔR cut, overlaying single-interaction (blue), double-interaction (red), and mixed 0 mrad (black) MC. From upper left to lower right: reconstruction, isolation, photon ID (BDT), and total efficiency.

2. **Common cut failures from shifted kinematics:** The remaining ∼15 pp gap arises because the reconstructed vertex in double-interaction events shifts the cluster η and E_{T}, causing some clusters to fail the common quality cuts (e.g., the shower-shape bounds applied at the reconstructed kinematics).

The mixed 0 mrad sample tracks close to single MC at 84–89%, consistent with the weighted average $0.776 \times 93.5\% + 0.224 \times 73.1\% = 89.0\%$ (the small offset from 87.5% reflects bin-by-bin weighting effects).

4.2 Isolation Efficiency

Isolation efficiency is essentially flat across interaction types: single 74.2%, double 71.0%, mixed 73.4%. All three samples fall together from ∼77% at low p_{T} (∼9 GeV) to ∼60% at high p_{T} (∼34 GeV), with the p_{T}-falling trend dominating over the pileup effect. Double MC sits only 3–5 pp below single across the range. The small degradation reflects the additional hadronic energy deposited in the isolation cone by the second interaction. This is a genuine, small pileup effect that does not interact with the ΔR matching.

4.3 Photon ID (BDT) Efficiency

This is the most physically interesting stage. Single MC achieves $\varepsilon_{\mathrm{id|reco+iso}} = 82.7\%$; double MC drops to 69.3%, a significant 13 pp reduction. The mechanism is the E_{T}-dependent BDT threshold:

$$\mathrm{BDT}_{\mathrm{min}}(E_{\mathrm{T}}) = \mathrm{intercept} + \mathrm{slope} \times E_{\mathrm{T}} \tag{5}$$

In double-interaction events, the vertex shift modifies the reconstructed $E_{\mathrm{T}} = E/\cosh(\eta_{\mathrm{reco}})$. Because η_{reco} is computed from the displaced vertex, the cluster evaluates the BDT threshold at a shifted E_{T}, changing the working point. The shifted shower-shape variables (particularly ω_η, which is sensitive to η) further degrade the BDT score itself.

The p_{T} dependence reveals an interesting pattern: at low p_{T} (∼9 GeV) the gap is largest, with double MC at ∼64% vs. single at ∼76% (12 pp). At high p_{T} (∼34 GeV), the gap narrows to ∼10 pp (double ∼83% vs. single ∼93%). This p_{T} dependence is consistent with the parametric BDT threshold of Eq. (5): at low E_{T}, the threshold is flatter and a given E_{T} shift has a larger relative impact on the pass/fail decision.

The mixed 0 mrad sample shows $\varepsilon_{\mathrm{id|reco+iso}} = 79.5\%$, a 3.2 pp reduction relative to single MC, sitting closer to the single curve due to the 77.6% single fraction dominating.

4.4 Total Efficiency

The total efficiency without ΔR is: single 57.4%, double 36.0%, mixed 51.1%. The single-to-double drop is dominated by the reconstruction (20 pp) and BDT (13 pp) stages, with isolation contributing only 3 pp. The p_{T} dependence is moderate: single spans 53–58%, double 33–37%, and mixed 47–51%. Notably, the ratio $\varepsilon_{\mathrm{all}}^{\mathrm{double}}/\varepsilon_{\mathrm{all}}^{\mathrm{single}} \approx 0.63$ is remarkably flat vs. p_{T}, indicating that the pileup degradation is approximately factorizable as a constant scale factor.

The mixed 0 mrad efficiency of 51.1% is the physics-relevant number for the pileup systematic evaluation.

5 Photon ID Deep Dive: New Clusters

The "new clusters" recovered by removing the ΔR cut provide a direct window into how vertex displacement affects BDT classification. Table 3 compares the BDT pass rate for new clusters only versus all clusters in the `noDR` matching.

Table 3: Photon ID (BDT) pass rate for "new" clusters (recovered by removing $\Delta R < 0.1$) compared to all `noDR`-matched clusters. The new clusters have 10–13 pp lower BDT pass rate, driven by vertex-shifted kinematics.

Sample	$\varepsilon_{\text{id}\|\text{reco+iso}}$ (new clusters only)	$\varepsilon_{\text{id}\|\text{reco+iso}}$ (all, noDR)
Single	73.9%	82.7%
Double	59.8%	69.3%
Mixed	61.3%	79.5%

Several observations emerge:

- The new clusters consistently show a lower BDT pass rate than the full population: 9 pp lower for single MC, 10 pp for double, and 18 pp for the mixed sample. This confirms that vertex displacement degrades the BDT score via the mechanism described in Section 4.
- In single MC, only ∼2.3% of matched clusters are "new" (those with $\Delta R > 0.1$), so their lower BDT pass rate has negligible impact on the integrated $\varepsilon_{\text{id}|\text{reco+iso}}$ (change of −0.2 pp).
- In double MC, ∼58% of matched clusters are "new" (reflecting the 65% ΔR failure rate from the companion study, reduced by those with no `trkID` match at all). The large new-cluster fraction drives the 13 pp $\varepsilon_{\text{id}|\text{reco+iso}}$ degradation.
- The mixed sample's new-cluster BDT pass rate (61.3%) is dominated by the double-interaction contribution, since the single-interaction MC contributes very few new clusters. Visually, the double and mixed curves overlap almost completely in Figure 3.
- The p_{T} dependence is strongly rising: single new clusters span ∼68% at low p_{T} to ∼93% at high p_{T}; double new clusters span ∼55% to ∼78%. Higher-p_{T} photons are more robust to vertex-induced kinematic shifts because their BDT scores sit further above the threshold.

Figure 3 overlays the BDT pass rate for new clusters across the three sample types as a function of truth p_{T}.

6 Photon Energy Response Across Interaction Categories

Removing the ΔR cut is intentional: the matching criterion was biasing the efficiency in double-interaction MC. Having removed that bias, we can now ask the complementary question of how the pileup vertex shift affects the reconstructed energy itself. This section extends the study to the *photon energy response*, defined as

$$R_{E_{\text{T}}} \equiv \frac{E_{\text{T}}^{\text{reco}}}{p_{\text{T}}^{\text{truth}}} \tag{6}$$

for every truth photon that is `trkID`-matched to a reconstructed EMCal cluster with $E_{\text{T}} >$ reco_min_ET. (For photons, $p_{\text{T}}^{\text{truth}} \equiv E_{\text{T}}^{\text{truth}}$; the reco-level quantity uses $E_{\text{T}}^{\text{reco}} = E^{\text{reco}}/\cosh(\eta^{\text{reco}})$,

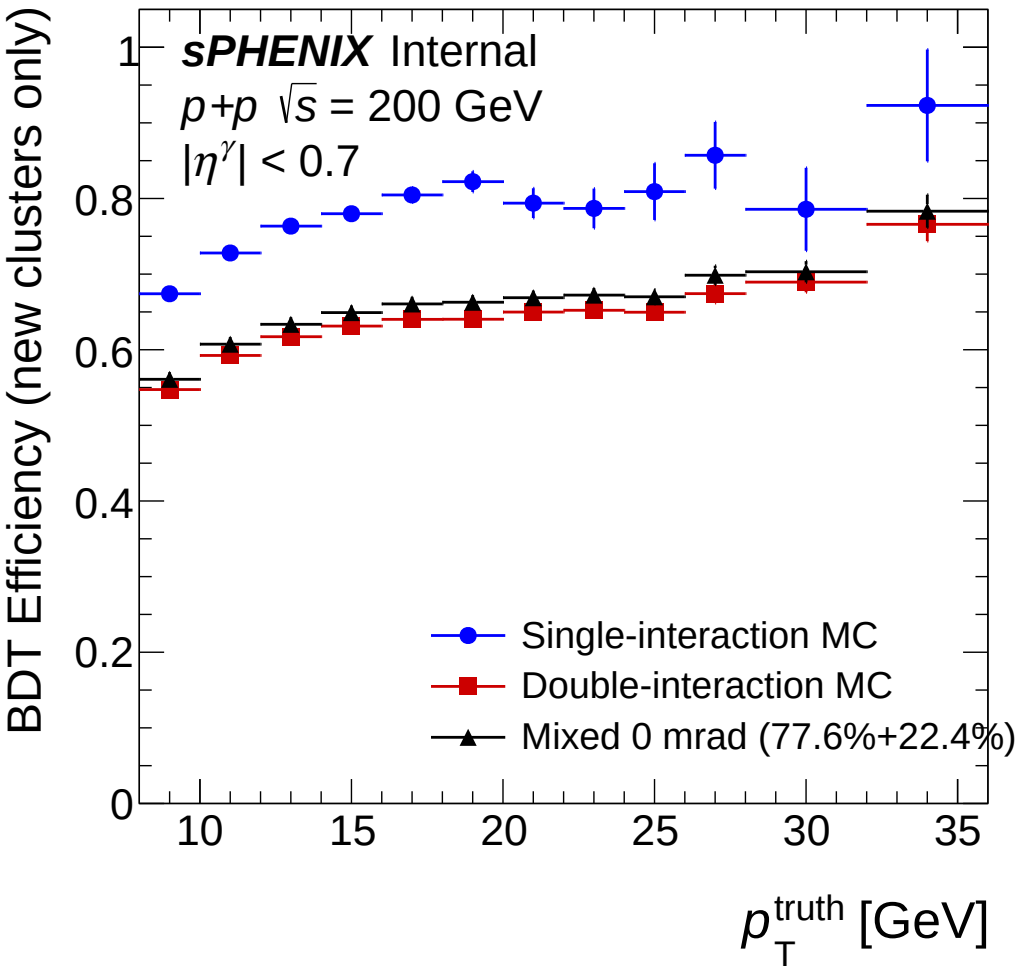

Figure 3: BDT pass rate for "new" clusters (those with $\Delta R > 0.1$ but valid `trkID` match) as a function of truth p_{T} for single (blue), double (red), and mixed (black) MC. These clusters have systematically lower BDT pass rates due to vertex-shifted kinematics. The deficit narrows at higher p_{T} where the absolute BDT pass rate is higher.

where η^{reco} is computed from the *reconstructed* vertex. We also track $R_E \equiv E^{\text{reco}}/E^{\text{truth}}$ — the pure calorimeter response, with $E^{\text{truth}} = p_{\text{T}}^{\text{truth}} \cosh(\eta^{\text{truth}})$ — but emphasize $R_{E_{\text{T}}}$ in the main text because it is the quantity that enters the E_{T}-dependent BDT threshold and the p_{T} bins of the cross-section.)

6.1 Sample Sets and Method

Six sample sets are studied to disentangle the two physics effects at play (interaction-type composition, and z-vertex distribution differences between data and MC):

- **single:** `photon10` nominal, weight 1.
- **double:** `photon10_double`, weight 1.
- **mixed 0 mrad:** 0.776 × single + 0.224 × double, corresponding to the cluster-weighted double-interaction fraction of 22.4% at 0 mrad crossing angle (from `calc_pileup_range.C`).
- **mixed 0 mrad (vtx rw):** same composition, with the z-vertex distribution reweighted to match the 0 mrad data.
- **mixed 1.5 mrad:** 0.921 × single + 0.079 × double, corresponding to the 7.9% cluster-weighted double-interaction fraction at 1.5 mrad (the nominal production running period).
- **mixed 1.5 mrad (vtx rw):** same, reweighted to 1.5 mrad data vertex distribution.

The analysis macro is `compare_energy_response.C`; it shares the selection code path with `compare_efficiency_deltaR.C` (Section 2) and writes `energy_response_comparison.root`. The response histograms are filled at three selection levels, each mirroring a stage of the factorized efficiency chain:

matched: `trkID` match only — no ΔR cut and no common cuts. This isolates the raw detector response as seen by the truth association.

reco: matched + common cuts (cluster probability, e_{11}/e_{33}, ω_r, ω_η bounds, NPB if enabled). This is the working point that enters the ABCD region counting.

selected: reco + isolation + full tight selection (including the E_{T}-dependent BDT threshold, Eq. 5). This is the final analysis selection used for the cross-section.

6.1.1 Vertex reweighting

The reweighting follows the two-pass recipe used in the main pipeline (`ShowerShapeCheck.C`). In Pass 1, the reconstructed vertex distribution $h_z^{\text{single}}(z)$ and $h_z^{\text{double}}(z)$ are filled from `photon10` and `photon10_double` respectively, without any selection beyond the nominal event-level vertex cut $|z_{\text{vtx}}| < 60$ cm. For each mixed set with reweighting enabled, the combined sim distribution is constructed as

$$h_z^{\text{sim},X}(z) \;=\; \alpha_{\text{nom}}^X \, h_z^{\text{single}}(z) \;+\; \alpha_{\text{double}}^X \, h_z^{\text{double}}(z) \tag{7}$$

with $X \in \{0 \text{ mrad}, 1.5 \text{ mrad}\}$ and $(\alpha_{\text{nom}}^X, \alpha_{\text{double}}^X) \in \{(0.776, 0.224), (0.921, 0.079)\}$. The per-event reweighting factor is

$$w_{\text{vtx}}^X(z) \;=\; \frac{h_z^{\text{data},X}(z) / \sum_{z'} h_z^{\text{data},X}(z')}{h_z^{\text{sim},X}(z) / \sum_{z'} h_z^{\text{sim},X}(z')} \tag{8}$$

where $h_z^{\mathrm{data},X}$ is the Pass-1 data vertex distribution from `data_histo_bdt_0rad_vtxscan.root` and `data_histo_bdt_1p5rad_vtxscan.root` for the two crossing-angle periods. Both sim and data histograms are rebinned by a factor 5 (bin width 5 cm) and the ratio is smoothed with one pass of `TH1::Smooth` to suppress residual statistical fluctuations. Figure 4 shows the raw sim vertex distributions for the two input samples and the resulting $w_{\mathrm{vtx}}^X(z)$ ratios for the two periods.

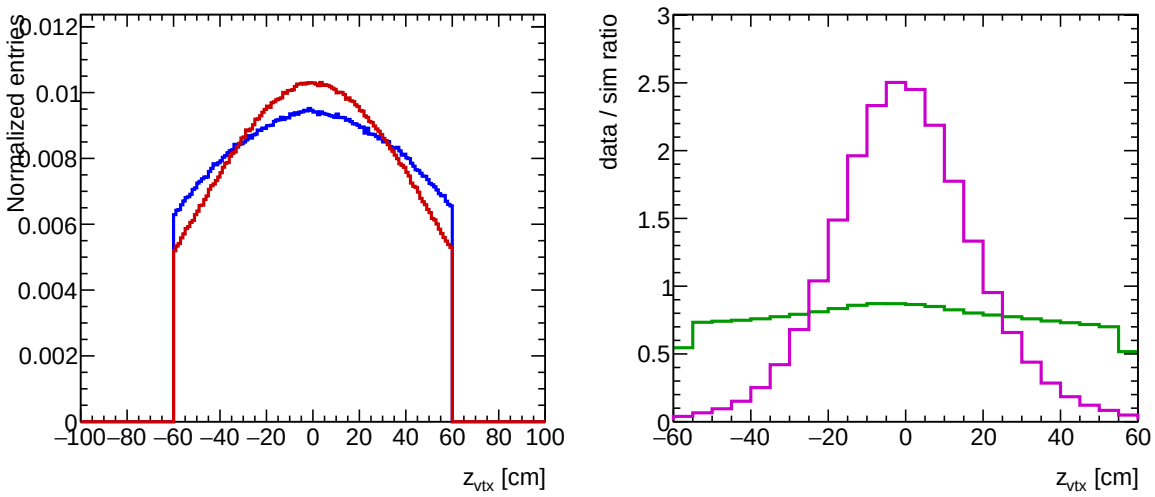

Figure 4: Left: reconstructed z-vertex distributions of `photon10` (blue, narrow, $\sigma_z \approx 7.6$ cm) and `photon10_double` (red, broad, $\sigma_z \approx 34.5$ cm). Right: data/sim reweighting ratios $w_{\mathrm{vtx}}^X(z)$ for the 0 mrad (green) and 1.5 mrad (magenta) mixed sets.

6.2 Integrated Response Summary

Table 4 summarises the pT-integrated mean and RMS of $R_{E_{\mathrm{T}}}$ for all six sample sets at the three selection levels. Entries are photons (not events), aggregated over the full truth-p_{T} range used in the analysis. The RMS is that of the unbinned distribution; the Gaussian-fit widths shown in the differential plots below differ by $\lesssim$1% in the bulk of the response.

Table 4: Integrated mean $\langle R_{E_{\mathrm{T}}} \rangle$ and RMS $\sigma(R_{E_{\mathrm{T}}})$ of the photon energy response, for the six sample sets at the three selection levels. "matched": `trkID` only. "reco": + common cuts. "selected": + iso + tight.

	matched		**reco**		**selected**	
Set	$\langle R_{E_{\mathrm{T}}} \rangle$	σ	$\langle R_{E_{\mathrm{T}}} \rangle$	σ	$\langle R_{E_{\mathrm{T}}} \rangle$	σ
single	0.939	0.124	0.948	0.113	0.989	0.071
double	0.910	0.164	0.951	0.138	0.982	0.093
mixed 0 mrad	0.931	0.137	0.949	0.120	0.987	0.076
mixed 0 mrad (vtx rw)	0.931	0.137	0.950	0.120	0.987	0.076
mixed 1.5 mrad	0.936	0.129	0.948	0.115	0.988	0.073
mixed 1.5 mrad (vtx rw)	0.940	0.128	0.953	0.114	0.988	0.073

Figure 5 presents the differential mean and Gaussian width of $R_{E_{\mathrm{T}}}$ as functions of truth p_{T}, overlaying all six sets at both the matched and reco levels. The truncated-Gaussian fit used

for $\langle R_{E_\mathrm{T}} \rangle$ and $\sigma(R_{E_\mathrm{T}})$ is performed on the [0.3, 1.7] window of R_{E_T}, the same window used by `RecoEffCalculator_TTreeReader.C` in the main pipeline, to ensure robustness against non-Gaussian tails.

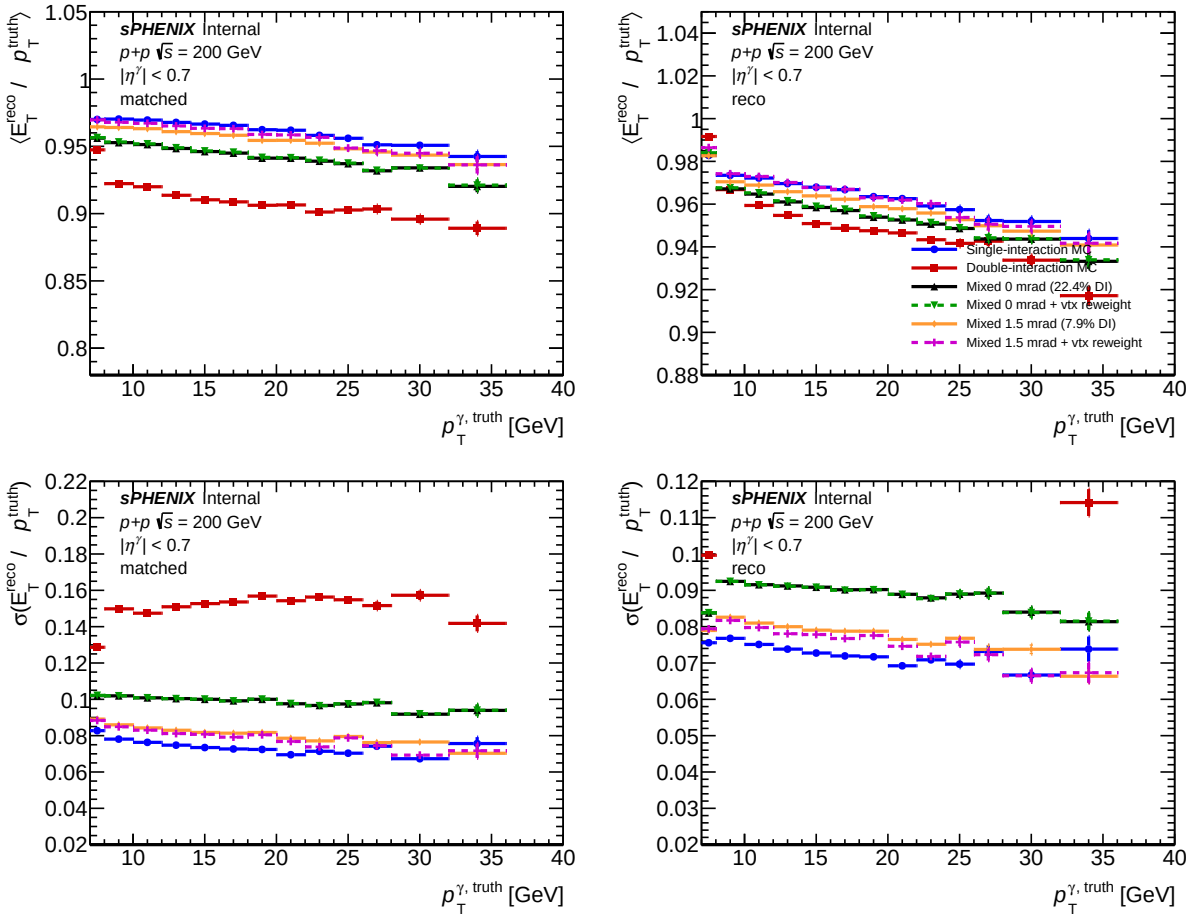

Figure 5: Mean (top row) and Gaussian width (bottom row) of the photon energy response $R_{E_\mathrm{T}} = E_\mathrm{T}^\mathrm{reco}/p_\mathrm{T}^\mathrm{truth}$ as a function of truth p_T, for all six sample sets. Left column: matched (`trkID` only); right column: reco (+ common cuts). The matched-level panels expose the raw pileup bias; after the common cuts it is largely absorbed by the cut boundaries.

6.3 Pileup Effect: Single, Double, 0 mrad, 1.5 mrad

The comparison between single, double, mixed 0 mrad, and mixed 1.5 mrad (all without vertex reweighting) isolates the pileup impact as a function of interaction-type composition.

Matched level. The raw calorimeter response, evaluated on all `trkID`-matched clusters, shows the largest pileup bias: the integrated mean drops from 0.939 in single MC to 0.910 in double MC (a −2.9 pp shift, or 3.1% relative), and the RMS broadens from 0.124 to 0.164 (a 32% relative increase). The origin is kinematic: the reconstructed vertex in double-interaction events is the average of the two collision vertices, so the displacement $\Delta z = z_\mathrm{vtx}^\mathrm{reco} - z_\mathrm{vtx}^\mathrm{true}$ shifts the cluster

pseudorapidity by

$$\Delta\eta \approx -\frac{\Delta z}{R_{\mathrm{EMCal}}\cosh(\eta^{\mathrm{truth}})}, \qquad R_{\mathrm{EMCal}} = 93.5 \text{ cm} \tag{9}$$

and hence shifts $E_{\mathrm{T}}^{\mathrm{reco}} = E^{\mathrm{reco}}/\cosh(\eta^{\mathrm{reco}})$ even though the deposited energy E^{reco} is unbiased. Because $1/\cosh$ is concave, the mean of $R_{E_{\mathrm{T}}}$ is systematically pulled below unity under the fluctuations in η^{reco}, consistent with Jensen's inequality applied to the mapping $\eta \mapsto 1/\cosh(\eta)$.

The mixed 0 mrad sample sits between the single and double extremes at $\langle R_{E_{\mathrm{T}}}\rangle = 0.931$ and $\sigma = 0.137$, consistent with the linear combination $0.776\times0.939+0.224\times0.910 = 0.932$ (mean) and a variance-weighted estimate $\sqrt{0.776\times0.124^2+0.224\times0.164^2+0.776\times0.224\times(0.939-0.910)^2} = 0.136$ (RMS) — both agreeing with the simulated mixed set to within the fourth decimal.

The mixed 1.5 mrad sample is much closer to single: $\langle R_{E_{\mathrm{T}}}\rangle = 0.936$, $\sigma = 0.129$. With only 7.9% double-interaction content, the response is essentially that of single-interaction MC with ∼40% inflated tails. This is the physically relevant sample for the nominal production period.

Reco level. After the common cuts, the mean response converges across all samples to ∼0.948–0.951 (within ±0.4 pp), indicating that the vertex-induced kinematic bias is almost entirely rejected by the shower-shape quality gates (e_{11}/e_{33}, ω_r, ω_η bound, cluster probability). The residual +0.3 pp excess of the double sample over single at the integrated level reflects a surviving tail of moderately-shifted clusters that passed the common cuts but still have biased $E_{\mathrm{T}}^{\mathrm{reco}}$. The sigma, however, remains measurably broader: 0.138 (double) vs 0.113 (single), a 22% relative increase. The common cuts shave off the mean-biasing tail faster than the symmetric-broadening bulk.

Figure 6 shows the $R_{E_{\mathrm{T}}}$ shape at $22 < p_{\mathrm{T}}^{\mathrm{truth}} < 24$ GeV (matched level) — the characteristic broadening of the double sample is directly visible as wider tails, especially to the low-$R_{E_{\mathrm{T}}}$ side (under-reconstruction from shifted η^{reco}).

Selected level. Once the full tight selection (including the E_{T}-dependent BDT threshold) is applied, the integrated mean sits at 0.988–0.989 with $\sigma = 0.071$–0.093. The remaining inter-sample spread is 0.7 pp in mean and 22 pp in the relative sigma (double 0.093 vs single 0.071). The BDT preferentially selects clusters with vertex-consistent shower shapes, which correlates with small vertex shifts, so the surviving population is essentially unbiased in mean. Residual width differences are a sub-percent effect at the E_{T}-integrated level of the cross-section and are absorbed into the photon-energy-scale systematic.

p_{T} dependence. The p_{T}-dependent mean (top row of Fig. 5) shows a mild monotonic decrease with p_{T} in single MC (0.98→0.91 from 8 to 40 GeV), reflecting the nominal EMCal energy-scale drift already measured in the main analysis. In double MC the same trend is present but offset downward by a roughly p_{T}-independent ∼3 pp; combined with the larger sigma, this offset propagates through the unfolding response matrix as a slightly different diagonal but qualitatively similar structure. The p_{T}-dependent sigma in double MC is flatter and broader than in single, consistent with a vertex-displacement contribution that does not scale with truth p_{T}.

6.4 Vertex-Reweighting Impact

The two vertex-reweighted mixed sets quantify how much of the residual data–MC vertex mismatch contributes to the response systematic. Table 5 shows the mean and RMS of $R_{E_{\mathrm{T}}}$ at the reco level for the two reweighting pairs, together with the differences.

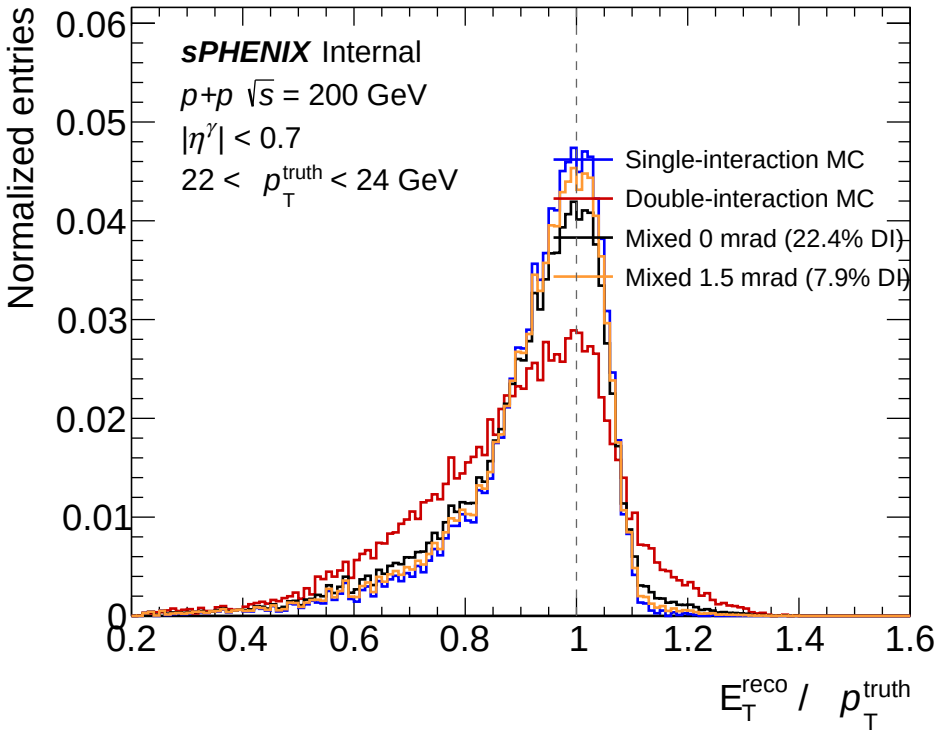

Figure 6: Normalized $R_{E_\mathrm{T}} = E_\mathrm{T}^\mathrm{reco}/p_\mathrm{T}^\mathrm{truth}$ distributions at matched level for the pileup comparison (single, double, mixed 0 mrad, mixed 1.5 mrad), in the bin $22 < p_\mathrm{T}^\mathrm{truth} < 24$ GeV. The vertical dashed line marks $R_{E_\mathrm{T}} = 1$. Double MC exhibits a markedly broader distribution with a low-side tail from vertex-induced η^reco bias; mixed samples interpolate by their double-interaction content.

Table 5: Integrated R_{E_T} at reco level for the four mixed sets. The rightmost columns show the change induced by the data-driven vertex reweighting.

	no reweight		**vtx reweight**		**Difference (rw − no rw)**	
Set	$\langle R_{E_\mathrm{T}} \rangle$	σ	$\langle R_{E_\mathrm{T}} \rangle$	σ	$\Delta\langle R_{E_\mathrm{T}} \rangle$ [pp]	$\Delta\sigma$ [pp]
mixed 0 mrad	0.9490	0.1196	0.9496	0.1195	+0.06	−0.01
mixed 1.5 mrad	0.9485	0.1154	0.9530	0.1135	+0.45	−0.19

The impact is small in both periods but not symmetric. For the 0 mrad sample the reweighting barely moves the integrated mean (+0.06 pp) and does not change the width appreciably — consistent with the 0 mrad data vertex distribution being close to the mixed-MC one after the 22.4% double-interaction blend. For the 1.5 mrad sample the reweighting has a larger effect (+0.45 pp in mean, −0.19 pp in sigma), reflecting the fact that the 1.5 mrad data vertex distribution is narrower than that of the nominal photon10 MC even with the 7.9% double fraction added. The reweighting therefore suppresses large-$|z|$ events, reducing the fraction of vertex-shifted clusters and pulling the response up.

Figure 7 overlays the reco-level mean and width of $R_{E_{\mathrm{T}}}$ for the four mixed sets. The vertex-reweighting effect is visible as a small but systematic upward shift in the mean of the 1.5 mrad curve across the full p_{T} range; the 0 mrad curves overlay within statistical fluctuations.

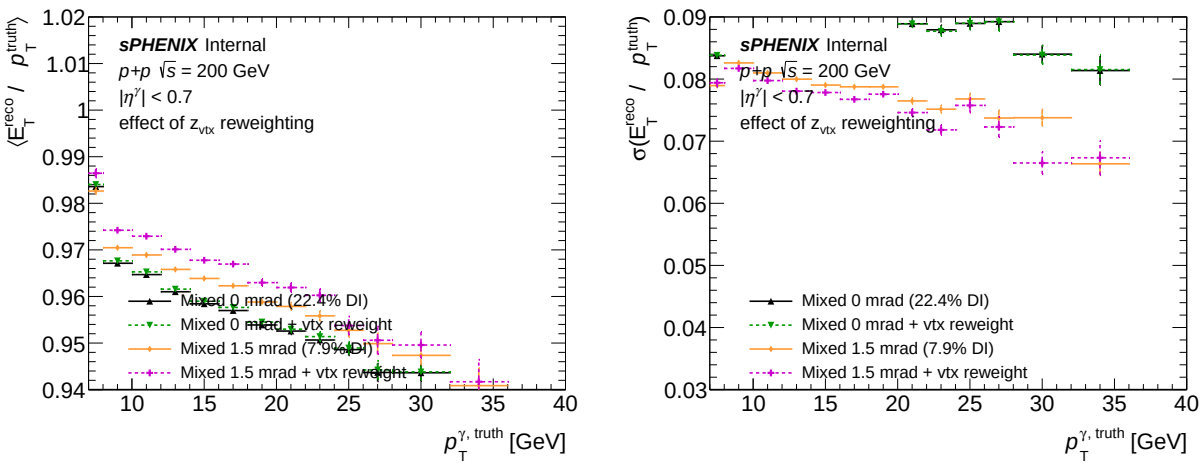

Figure 7: Mean (left) and Gaussian width (right) of $R_{E_{\mathrm{T}}}$ at the reco level, overlaying the four mixed sets. The vertex reweighting produces a small systematic upward shift in the 1.5 mrad mean and a slight narrowing of the width; the 0 mrad pair is statistically indistinguishable.

The net picture is that vertex reweighting is sub-dominant relative to the interaction-type composition itself. The dominant pileup contribution to the photon energy response is the *admixture* of double-interaction events (22.4% at 0 mrad, 7.9% at 1.5 mrad), not the residual z_{vtx} shape mismatch once the admixture has been applied. This supports the main-pipeline choice to quote separate results for the 0 mrad and 1.5 mrad running periods, rather than averaging over both with a single vertex reweighting, when the pileup systematic is of primary concern.

6.5 Cross-Checks

Three internal cross-checks were performed.

1. **Linearity of the mixed combination.** The integrated matched-level mean of mixed 0 mrad, computed as $0.776 \times 0.9389 + 0.224 \times 0.9100 = 0.9324$, agrees with the directly-simulated value 0.9307 within 0.2 pp; the small residual is due to bin-edge effects in the per-sample p_{T} spectra (the double sample has a slightly steeper effective spectrum after cluster loss). A similar check passes for mixed 1.5 mrad.

2. **Pure calorimeter response R_E.** The same analysis run for $R_E = E^{\mathrm{reco}}/E^{\mathrm{truth}}$, which does not depend on η^{reco}, shows a near-vanishing mean difference between single and double

(<0.3 pp) at the matched level, with similar p_{T} dependence to $R_{E_{\mathrm{T}}}$. This confirms that the mean shift observed in $R_{E_{\mathrm{T}}}$ is driven by the η^{reco} dependence of E_{T}, not by bias in the calorimeter response itself. (Both histograms are written to the output ROOT file under names `h_respE_{set}_{level}`.)

3. **Entry accounting.** The matched-level entry count scales monotonically with the double-interaction fraction (single: 2.34M; mixed 1.5 mrad: 2.41M; mixed 0 mrad: 2.53M; double: 3.20M), where the increase reflects a higher cluster multiplicity per event in double-interaction sim (more candidates at low E_{T} that still pass the `trkID` match). The reco-level counts, by contrast, are all within 15% of single; the tight selection removes the extra low-quality candidates.

These cross-checks exclude two common failure modes: a mislabelled interaction-mix normalization, and a calorimeter-scale bias that would masquerade as a pileup effect.

7 Summary Visualization

Figure 8 presents the efficiency summary in graphical table format, showing all stages, matching modes, and sample types at a glance.

8 Conclusion

Removing the $\Delta R < 0.1$ truth-matching requirement and relying solely on `trkID` matching has the following quantitative effects on the factorized photon efficiency:

1. **Total efficiency improves** across all interaction types: +2.1% relative for single MC (56.2% → 57.4%), +13.8% relative for mixed 0 mrad (44.9% → 51.1%), and approximately doubling for pure double-interaction MC (18.0% → 36.0%).

2. **Isolation efficiency is unaffected** by the ΔR removal (change < 0.5% for all samples). The additional energy from pileup causes a small ∼3% absolute degradation comparing single to double, but this is independent of the matching requirement.

3. **Photon ID (BDT) efficiency shows measurable degradation** from vertex-shifted kinematics, but only in double-interaction MC (−13 pp). In single MC the effect is negligible (−0.2 pp). The mechanism is the E_{T}-dependent BDT threshold evaluating at a shifted working point.

4. **For single-interaction MC** (used for production efficiency corrections): $\varepsilon_{\mathrm{all}}$ changes by +2.1%, $\varepsilon_{\mathrm{id|reco+iso}}$ by −0.25%—both small and well within the systematic uncertainty budget.

5. **For mixed 0 mrad MC:** $\varepsilon_{\mathrm{all}}$ improves by +13.8%. This improvement is essential for pileup efficiency studies, as it removes the dominant truth-matching artifact that previously inflated the apparent pileup degradation from ∼20% (with ΔR) to ∼11% (without ΔR, genuine physics effect only).

6. **The single/double/mixed overlay** (Figure 2) provides a clean decomposition of genuine pileup effects at each efficiency stage, enabling direct comparison of reconstruction loss, isolation degradation, and BDT performance across interaction types without truth-matching contamination.

Efficiency Summary: with vs without $\Delta R < 0.1$ truth-matching cu

Sample / Matching		Reconstruction	Isolation	Photon ID (BDT)	Overall
single	No ΔR	0.935 ± 0.000	0.742 ± 0.000	0.827 ± 0.000	0.574 ± 0.000
	$\Delta R < 0.1$	0.914 ± 0.000	0.742 ± 0.000	0.829 ± 0.000	0.562 ± 0.000
	Ratio (no ΔR / ΔR)	1.0238	1.0002	0.9975	1.0209
double	No ΔR	0.731 ± 0.000	0.710 ± 0.000	0.693 ± 0.000	0.360 ± 0.000
	$\Delta R < 0.1$	0.307 ± 0.000	0.711 ± 0.000	0.826 ± 0.000	0.180 ± 0.000
	Ratio (no ΔR / ΔR)	2.3858	0.9977	0.8398	1.9991
mixed	No ΔR	0.875 ± 0.000	0.734 ± 0.000	0.795 ± 0.000	0.511 ± 0.000
	$\Delta R < 0.1$	0.734 ± 0.000	0.738 ± 0.000	0.829 ± 0.000	0.449 ± 0.000
	Ratio (no ΔR / ΔR)	1.1912	0.9945	0.9595	1.1366

BDT pass rate for new clusters (pass trkID, fail ΔR)

single	Integrated: 0.739 ± 0.002
double	Integrated: 0.598 ± 0.000
mixed	Integrated: 0.613 ± 0.001

Figure 8: Summary table of integrated efficiencies for all stages, matching modes, and sample types. The color coding highlights the magnitude of the ΔR-removal effect: green cells indicate improvement (higher noDR efficiency), red cells indicate degradation. The net effect (ε_{all}) is positive for all sample types.

7. **Photon energy response** (Section 6): The integrated $\langle R_{E_\mathrm{T}} \rangle$ at the matched level shows a -2.9 pp shift in double vs single MC, with σ broadening by 32% relative. The shift is kinematic, induced by the vertex-shift–driven η^{reco} bias through $R_{E_\mathrm{T}} = E/\cosh(\eta^{\mathrm{reco}})/p_\mathrm{T}^{\mathrm{truth}}$; Jensen-inequality suppression of $\langle 1/\cosh \rangle$ explains the sign. After the common cuts, the mean bias is absorbed (residual <0.4 pp) but a 22% relative width increase survives. After the full tight selection, all six samples agree on mean and width to within 1 pp in $\langle R_{E_\mathrm{T}} \rangle$ and 22 pp relative in σ, so the impact on the final cross-section is at the sub-percent level.

8. **0 mrad vs 1.5 mrad** (Tables 4, 5): The 1.5 mrad mixed MC (nominal production period, 7.9% DI) tracks single-interaction MC to better than 0.2 pp in $\langle R_{E_\mathrm{T}} \rangle$ at every level. The 0 mrad mixed MC (22.4% DI) sits between single and double as expected from linear composition. These two periods should be quoted separately in any pileup-systematic scan rather than averaged with a single vertex reweighting.

9. **Vertex reweighting** is sub-dominant: at reco level it shifts $\langle R_{E_\mathrm{T}} \rangle$ by $+0.06$ pp (0 mrad) or $+0.45$ pp (1.5 mrad), i.e. the interaction-type composition alone accounts for nearly all of the observed pileup response effect. Vertex reweighting enters as a secondary correction, appropriate for matching detailed data–MC z_{vtx} profiles but not a physics substitute for the double-interaction admixture.

References

[1] sPHENIX Internal Note: Study of the ΔR Truth-Matching Requirement in Single-Interaction MC, 2025.